



ÖZYEĞİN UNIVERSITY
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SENIOR PROJECT REPORT

PARKOMFY

By

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Abstract

PARKOMFY offers innovative solution to the parking problem. In project scope using with existing security cameras over a YOLO algorithm real time detect parking slot occupancy. YOLO also detects vehicle license plates and system reads with EasyOCR then writes on SQL database with the timestamp. Mobile app allows user both see available parking spaces and pay of the parking fees.

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I. Introduction

Rapid urbanization and increasing vehicle numbers bring with the difficulty of finding parking slots which is the biggest problem in metropolises. Not only spending time searching for a suitable parking slot is a source of stress but also environmental problem due to unnecessary fuel consumption leads to carbon emission.

Traditional parking management systems generally rely on expensive sensors and can't be integrated into every area because of set up and maintenance costs.

This project, "PARKOMFY", digitalizes parking management by using computer vision and artificial intelligence. PARKOMFY aims to detect occupancy in parking slots by existing security cameras without any needs of extra sensor installation. Core of the project is detecting existence of vehicle in parking slots by using YOLO (You Only Look Once) and acquired data will store on SQL based database and provide to user real time information about parking slots via mobile application.

Report Roadmap

II. Background: Provides theoretical and technical foundations of the technologies in the project.

III. Problem Statement: Provides the main problem, scope of project and engineering requirements.

IV. Solution Approach: Provides the architecture of system, design decisions and implementation details.

V. Results and Discussion: Provides performance metrics, test results and analysis of system effectiveness.

VI. Related Work: Compares this project with existing solutions.

VII. Conclusion and Future Work: Summarizes project and discusses potential future improvements

Acknowledgements: Provides special thanks to contributors and supporters.

References: Provides information about cited sources

II. Background

Introduce the set of techniques, tools, technologies that are utilized in your project.

This part introduces main technologies and methodologies which used on development of the PARKOMFY system. The project brings together computer vision, database management and developing mobile application for the purpose of a seamless smart parking solution.

2.1. Computer Vision

OpenCV Library: Open-source library which used for tasks like image resize, crop and improve detection accuracy.

YOLO: Trained to detect occupancy of parking space and detect license plates for pass to EasyOCR.

EasyOCR: Used for read license plate.

2.2. Database Management System (SQL)

MySQL: SQL based database that is structured to manage data produced by parking system is used (entrance-leave time, customer id-plate, payment).

2.3 Mobile Application Frameworks

React Native: The UI of PARKOMFY is built with cross-platform mobile development framework React Native.

RESTful API: Mobile application ensures pull real-time parking data from database to communicating with backend server via RESTful APIs.

Stripe API: PARKOMFY uses stripe to take payments.one

III. Problem Statement

This section details the main problems that PARKOMFY aims to solve, scope of the project and possible assumptions and constraints.

3.1. Problem Formulation

Rapid increase of vehicle numbers has made the efficient use of existing parking capacity compulsory. Drivers spend time searching park spaces leads both waste of time and carbon emissions. Existing solutions requirements deploy physical sensors (ultrasonic or magnetic) for all spaces. This approach costly to install and hard to maintain that inefficient. PARKOMFY aims by the using existing security cameras with a software-based image processing.

3.2. Project Scope

PARKOMFY project that check slot occupancy by using object detection and reads the license plates with computer vision park in and out times stores data in cloud-based system and deliver to end user by a mobile application and provides a payment method as digital. It is not a part of any other project it is designed as a new project from scratch.

3.3 Engineering Problem

The fundamental engineering problem in PARKOMFY project is integrating an accurate YOLO model. Optimizing the model with minimum latency updates database and establishing seamless data synchronizations between the mobile application is the main engineering challenge of the project.

3.4 Assumptions and Constraints

-Assumptions

- Cameras assumed has good angles and clear vision and stable.

- It is assumed that system has seamless and stable internet.

-Constraints

- System performance limited by blind angle and low light resources (if camera has not night mode).

- Image processing performance limited by the GPU used.

3.5. Requirements

3.5.1. Functional Requirements

Vehicle Detection: System should understand the vehicle entrance, leaving time and link with the license plate for mobile payment.

Visual Map: Mobile app should provide floor plan and available spots.

Park Occupancy: The system must detect existence of vehicle in specified area with an accuracy of nearly 100%.

Real-Time Update: If occupancy changes in a park slot system should update information in both database and mobile app in a couple seconds.

Data Storage: Data occupancy information and entrance and leave time comparison to evaluate park price needs to be hold in database.

3.5.2. Non-Functional Requirements

Usability: Mobile interface should be easily understandable and user friendly.

Reliability: System should be active 7/24 and if any interruption occurs system must automatically reactivate itself.

Scalability: The system should be able to process data from multiple cameras.

Privacy: Personal data should be encrypted such as license plate to protect personal data.

IV. Solution Approach

This section contains PARKOMFY's technical architecture, technical problems faced during implementation and feasibility analyses.

4.1. Solution approach and Design

PARKOMFY contains 4 main components from check in to payment in parking experience.

1.LPR (License Plate Recognition): Camera on the entrance of parking area reads license plates of entered vehicle. This information saves to database with entrance time.

2.Object Detection (YOLO): Cameras inside the parking area continuously scan slots by using YOLO algorithm it detects whether parking slot empty or not.

3.Database and Pricing: Entrance time, license plate information and stay time in park holds in SQL database and when a car wants to check out system automatically calculates the fee according to pricing policy.

4.User Interface and Digital Payment: On mobile app users can see empty slots easily parks these slots and when leaving parking area they can pay with integrated payment method.

4.2. Design Decisions, Tools and Techniques

	Selected Tool	Advantages	Disadvantages
LPR	OCR	No physical ticket need, no paper waste	Some vehicles license plates might be unreadable due to mud or something else or they may not have any license plate
Object Detection	YOLO(YOLOv8)	Fast (works in real time), high accuracy	Dependent on training data and accuracy might decrease in low light
Database	SQL (MySQL)	Fast in relational queries and ensures data integrity	Can be challenging large data flows
Mobile Interface	React Native	Cross platform support (ios, android) fast and easy UI design development	Not perfect as real native apps (design problems – performance problems)
Payment Method	Stripe API	User without any extra physical effort can pay with mobile app directly – reduces traffic in the parking area exit	Commissions

Why image processing instead of physical sensors? Because a sensor can only detect one spot, but a camera can detect entire area it can see.

4.3. Implementation Issues

- Some big vehicles can block other park slot if the camera angle not good.
- License plate recognition sometimes makes up random letter or number.

4.4. Feasibility Analysis

Technical Feasibility: YOLOv8 algorithm, provides high accuracy and real time processing power. SQL based database architecture provides high speed of processing and holds entrance and leave times and payment records. Payment APIs such as Stripe or Iyzico allows financial transactions in secure. Open-source libraries wide support provides us convenience to develop the project within the right framework.

Operational Feasibility: System can use existing camera infrastructure so won't need any extra physical changes. Entrance and exit ticketing and payment like needs to human power works needless in the system. This situation makes parking management fully autonomous minimizes operational problems and reduces waiting times for users provides smoother flow.

Financial Feasibility: Traditional smart parking systems needs to magnetic or ultrasonic sensor set up costs for each slot, but PARKOMFY is a software base solution so lowers the set-up costs, and these sensors might need maintenance or repair over time it results extra cost. Our solution only need software updates no need any extra.

4.5 Utilized Knowledge

CS202 (Database Management Systems): SQL database schemas such as vehicle entrance and exit time, license plate records implemented with this course achievements.

CS320 (Software Engineering): The projects requirement analysis and structure of the system planning were developed according to this course achievements.

4.6. Sustainable Development Goals

SDG 9 (Industry, innovation and infrastructure): Instead of building new physical infrastructure PARKOMFY digitalize existing camera infrastructure with artificial intelligence and computer vision technologies it offers innovative solution that increase efficiency by transforms existing security cameras to smart sensors.

SDG 11 (Sustainable cities and communities): A portion of urban traffic is caused by vehicles which looking for parking spaces PARKOMFY reduces traffic congestion by directing drivers directly to available park spaces.

SDG 12 (Responsible consumption and production): As mentioned at 4.2. with autonomous license plate recognition prevent unnecessary paper consumption. This digitalizing process encourages more responsible consumption and enables operations conducted without paper.

SDG 13 (Climate action): Reduce on searching time for park spaces reduces also vehicles working time this leads to reduces fossil fuel consumption this directly reduces carbon emission and improves air quality.

V. Results and Discussion

In this part, there are particular research findings, how validating the result and possible bounders of the system.

5.1 Existing Findings and Literature Evaluation

The research yielded the following results after that research findings were done in the first stage of project and review of case studies:

Algorithm Selection: When similar smart parking applications were examined, it seems that YOLO architecture performs better than the other deep learning models for real-time vehicle detection.

Data Management: PARKOMFY needs a particular data integrity. The successful studies in literature mostly use relational databases to manage large scale parking data. Therefore, it is good to use relational databases in PARKOMFY.

5.2 Addressing the Problem and Validation Approach

The project is not in validation phase at this moment, so the validation of the project is deferred to CS402. However, the validation plan for the evaluate success of the system is created like in the following:

Error Analysis: The accuracy of model that will be developed will be validated by measuring the “True” and “False” precision recalls over test data sets.

Manual Comparison: The real-time occupancy data that the system generated will be compared manually with number of real vehicles in parking cameras. Therefore, reliability of the system will be tested.

5.3 Known Limitations

As a result of the theoretical researches done and design analysis, possible technic limitations which the system may be faced:

Environmental Factors: When case studies in literature were investigated, it seems that very low light levels or extreme weather conditions may affect object detection negatively.

Visual Range Limitations: Some vehicles may obscure themselves views, and this occlusion issue leads to that some parking lots are not detected, so this occlusion issue may be a serious limitation.

VI. Related Work

This section compares PARKOMFY and other Smart Parking Systems.

6.1. Sensor Based Systems

Traditional smart parking solutions rely on sensors physically placed on each park slots. This cause extra physical set up maintenance effort and costs according to our software-based system.

6.2. Other Computer Vision Solutions

Numerous image-processing based studies exist in literature, but most current studies stay only on detection stage with PARKOMFY end to end integration this detection will combine with license plate recognition and mobile payment method aims not only detecting faster and effective with YOLO but also provide whole parking management system.

VII. Conclusion and Future Work

7.1 Current Status and Key Achievements

In this CS401 stage, main design plan of the system and the literature research of PARKOMFY project are completed successfully. The scope of project has been identified, availability of YOLO architecture is confirmed for object detection, and communication architecture between database of the system and mobile application is designed in this process. Also, the engineering framework is made through describing functional and non-functional requirements of PARKOMFY project.

7.2 Remaining Tasks and Planning

These steps in below will be followed in order to implement the project completely in the next semester:

Data Gathering and Labeling (Partially Done): In order to train YOLO model, creating parking dataset that includes various weather and lightning conditions.

Model Training and Optimization (Started): Training the model for vehicle and empty lot detection, and making their validation tests.

Developing (Backend & Frontend): Creating SQL database, writing API endpoints and developing mobile application interface.

System Integration: Testing through connecting image processing unit and mobile application end to end.

7.3 Economic and Social Factors

It is expected that fuel consumption is reduced due to minimized driver's spending time that is sought for parking lots with adoption of PARKOMFY. Therefore, it will not ensure economic savings only in individual scope but also ensure that savings in national scope. For social perspective, reduction of traffic density will lower the stress of driver and it will also increase life quality in city.

7.4 Global Health, Environmental and Security Factors

Environmental and Health: The time that vehicles seek for empty lot will be reduced, so this will reduce carbon emissions directly and benefit for the global warming challenge.

Security and Legal Issues: There will be camera images and there must be managed legally with respect to the KVKK, so data security will be ensured with encrypted API connections.

7.6 Future Work

The following features are planned to be added in order to expand the scope of this project:

Reservation System: User can reserve for parking lots before came there.

Autonomous Vehicle Integration: Transferring smart parking data to the navigation systems of vehicles with no drivers.

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References

Donald C. Shoup, Cruising for parking, *Transport Policy*, Volume 13, Issue 6, 2006, Pages 479-486, ISSN 0967-070X, <https://doi.org/10.1016/j.tranpol.2006.05.005>.
(<https://www.sciencedirect.com/science/article/pii/S0967070X06000448>)

Hussain, M. (2023). YOLO-v1 to YOLO-v8, the Rise of YOLO and Its Complementary Nature toward Digital Manufacturing and Industrial Defect Detection. *Machines*, 11(7), 677.
<https://doi.org/10.3390/machines11070677>

Appendix